

# FIRST SEMESTER MCA

## 22-382-0107 PYTHON PROGRAMMING LAB

### LIST OF EXPERIMENTS

1. Write a program to demonstrate different number datatypes in python.
2. Write a program to perform different arithmetic operations on numbers in python
3. Write a program to create, concatenate and print a string and accessing substring from a given string.
4. Write a python script to print the current date in following format "Sun May 29 02:26:23 IST 2017"
5. Write a python program to create, append and remove lists in python.
6. Write a program to demonstrate working with tuples in python
7. Write a program to demonstrate working with dictionaries in python
8. Write a python program to find largest of three numbers
9. Write a python program to convert temperature to and from Celsius to Fahrenheit.
10. Write a python program to construct the following pattern using nested for loop:

```
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * *  
* * * *  
* * *  
* *  
*
```

11. Write a python program to print prim numbers less than 20
12. Write a python program to find factorial of a number using recursion.
13. Write a python program to define a module to find Fibonacci Numbers and import the module to another program.
14. Write a python program to define a module and import a specific function in that module to another program.
15. Write a script named copyfile.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file.
16. Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order.
17. Write a Python class to convert an integer to a roman numeral.
18. Write a Python class to implement pow(x, n)
19. Write a Python class to reverse a string word by word.

20. Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a given circle
21. Given a file "auto.csv" of automobile data with the fields index, company, body-style, wheelbase, length, engine-type, num-of cylinders, horsepower, average-mileage, and price, write Python codes using Pandas to
  1. Clean and Update the CSV file
  2. Print total cars of all companies
  3. Find the average mileage of all companies
  4. Find the highest priced car of all companies
22. Write a Python program to add two matrices and find the transpose of the resultant matrix.
23. Write a Python program to create a function called list\_of\_frequency that takes a string and prints the letters in non-increasing order of the frequency of their occurrences. Use dictionaries.
24. Write a Python program to read a list of numbers and sort the list in a nondecreasing order without using any built-in functions. Separate function should be written to sort the list wherein the name of the list is passed as the parameter.
25. Given the sales information of a supermarket as a CSV file with the following fields: month\_number, Soap, facewash, toothpaste, shampoo, moisturizer, total units, total profit.  
Write Python codes to visualize the data as follows
  - a) Toothpaste sales data of each month and show it using a scatter plot.
  - b) Soap and face wash product sales data and show it using the bar chart
  - c) Calculate total sale data for last year for each product and show it using a Pie chart.
26. Write a script named copyfile.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file. Write a program that inputs a text file. The program should print all the unique words in the file in alphabetical order.